

SHOP MEASUREMENT AND CALCULATIONS



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College

Shop Measurement and Calculations
Delivery Support Set for Instructors
By Robert Fergesen

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Licensing

A detailed breakdown of this resource's licensing can be found in [Back Matter/Detailed Licensing](#).

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1.1: Course Description

The HACC (Harrisburg Area Community College) catalog description for this course is:

Teaches students the basics of the measurement systems that are required for a variety of trades. This course focuses on such measuring systems as: tape measure usage, rulers, micrometers, calipers and a variety of electronic tools. In addition, this course covers basics of unit-of-measure conversions and the use of formulas and lookup tables for a variety shop calculations.

Context

This "book" in the LibreText tool was designed as a complete course delivery tool. The material within includes course supporting materials, such as PowerPoint slides, Test Question Libraries and Assignments. In addition it includes links to original content created for this course in the form of videos (links to YouTube) as well as external content on a wide variety of topics where great material existed.

Please note that this book is intended to be used by the instructor, and not for general access to all the students in its entirety. The instructor can certainly take as much content as needed from this book and make it available to her or his students, however, I would recommend removing access to the question banks.

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1.2: Background

Students entering the trades today come from a wide variety of backgrounds in math and basic measurement systems. Many programs require that trade students take a general math course as part of graduation requirements. This has lead to some serious challenges for some non-traditional students who may not have a strong math backgrounds. Many trades, though, do not require the broad skills found in general math courses. The trades require, more commonly, good measurement skills, unit of measure conversions, and a basic understanding of the application of formulas and lookup tables to generate the results they need in their work. This course was constructed to create a baseline math course that will provide simple techniques for students who need minimal, trades based, skills.

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1.3: Desired Learning Outcomes

- Identify and effectively use the correct measurement tool within an industrial environment, covering tape measures, rulers, micrometers, calipers, and a variety of electronic tools
 - Convert unit of measures from a variety of measurement systems including English to Metric, Inches to Feet to Yards, and other industrial measurement systems.
 - Evaluate word problems and select the correct formula or lookup table for a variety of basic industrial problems including mechanical systems, electrical, pneumatics, hydraulics, and sensor systems.
 - Use lookup tables for a variety of basic industrial problems.
 - Use formulas, use calculators, and manage units of measure for a variety of basic industrial calculations.t & Calculations.
-

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1.4: On Terminology

My Theory on learning terminology

When you learn the terms of a particular subject you start to develop a kind of superpower around that subject. By learning that language you begin the understanding. So by learning technology terms, and especially by going through and understanding what those terms mean you're really going to start to absorb what the manufacturing world, and the "mechatronics" world is all about.

For many the terminology seems a little bit less important than learning the content, but I think it's first and most important learning the terms first allows a deeper immediate understanding of the content.

Another aspect of learning terminology first is that if you meet somebody and you talk about their industry intelligently with the right terms you will make a positive impression. If you know the terms then you can speak in the language with anybody in that world. You will communicate and they are going to appreciate you more.

If you work with somebody and you prove to them you don't know what these key terms mean they won't appreciate you as much. This is a critical skill to get a job in mechatronics.

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1.5: Using the internet

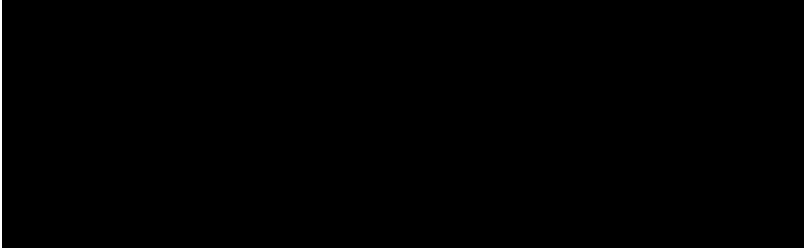
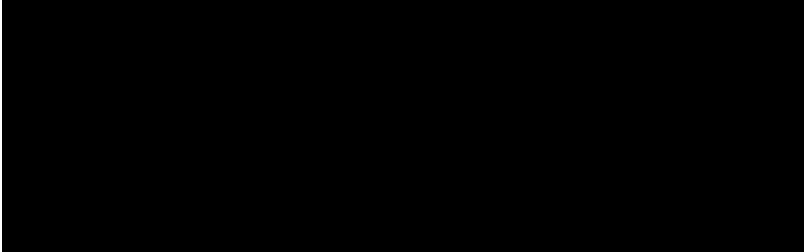
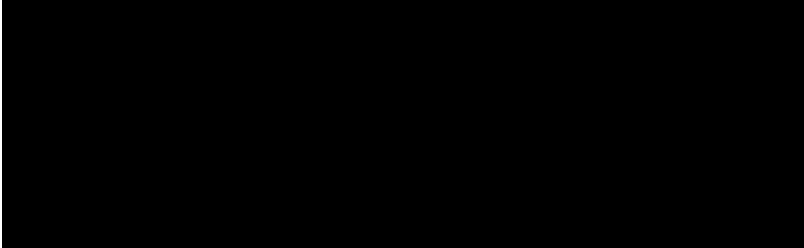
Encouraging students to cheat (using the internet)

To start off with I'm going to teach you guys number one you are allowed to cheat. You are allowed to cheat and cheat and cheat when it comes to using your cell phone and the internet, because in the job world most times you're going to have access to a computer or your cell phone.

Of course, I'm not saying cheat on tests, or use someone else's assignments, however, the ability to rapidly and effectively look up part information, troubleshooting experiences, datasheets, and a plethora of other information is a critical skill in any of the trades, especially Mechatronics.

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1.6: Video Links



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2.1: Linear Measurement

- Measurement Units: English vs Metric
- Simple Measurement Systems:
 - Rulers
 - Flexible tape measures
 - How to take a measurement with a tape or ruler
 - Determine the unit of measure.
 - Determine the fraction for each segments. This will be the denominator for the results
 - Sight down above ruler to workpiece being measured; angled viewing will skew the results
 - Round to the nearest line segment
 - Count the lines from the previous whole unit, and that is the numerator your fraction
 - Simplify the fraction if possible
 - Converting to decimal for drawings or some documentation

Video Links:

- [Measurement Systems Lecture Part 1](#)
- [Measurement Systems Lecture Part 2](#)
- [Measurement with Calipers](#)
- [Special Rulers Lecture](#)
- [Tape Measure Pro Tips](#)
- [Old school steel ruler video](#)
- [Old School Micrometer](#)
- [7 best measurement systems](#)
- [Micrometer Use Video](#)
- [Electronic Caliper Usage](#)
- [Introducing The Linear Measurement Assignment](#)

Reading Links:

- [Tape Measures Reviewed](#)
- [Example Electronic Measurement Systems](#)

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2.2: Temperature Measurement

- Laser Measurement
 - Room Measurement Systems
 - Long Distance Laser Measurement
 - Alignment Systems
- Ultrasonic Measurement System
 - Tank level
 - Position location
- Vision Location Systems

Video Links

- [Temperature Measurement Lecture Part 1](#)
- [Temperature Measurement Lecture Part 2](#)
- [Thermometer Basics](#)

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2.3: Pressure measurement

- Pressure Gauges
- Introducing PSI
- Dial Gauges
- Example of Pressure Control Automation

Video Links

- [Pipes & Pressure](#)
- [How a pressure gauge works](#)

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2.4: Other Measurement Systems

- Rate Measurement Systems
 - Velocity
 - Flow rate
- Vision systems
 - Color / Temperature
 - Scanning for positional errors

Video Links

- [Machine Vision Systems](#)
- [Flow Measurement](#)

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2.5: Converting between fractions and decimal notation

Video Links

- [Fractions to Decimals](#)
- [Decimals to Fractions](#)

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3.1: Rules for UOM

Rules For UOM Conversion



ALWAYS
Show the unit of measure in your calculations



Create A Multiplier
Create a conversion multiplier with the unit you want to change TO on the top (Numerator) and the unit you want to change FROM on the bottom (Denominator)



REALITY CHECK
Check your results

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3.2: Video Links



Unit Conversion the Easy Way (Dime Book)

A weightlifter can lift 495 lbs. How many kg is that?

1 kg = 2.2 lbs. (conversion factor)

$$495 \text{ lbs.} \left(\frac{1 \text{ kg}}{2.2 \text{ lbs.}} \right) = ? \text{ kg}$$

 495 ÷ 2.2 = 225

Watch on  YouTube

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4.1: Formula selection

The first step in effective use of formulas for the trades is finding good sources. I recommend becoming familiar with the Google, which can be used to find most common formula and basic engineering. In addition, I recommend the use of common (hardcopy) reference books, such as Pocket REF by Thomas J. Glover.

Videos:

[Selecting The Formula](#)

[Triangle Method Lecture](#)

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4.2: Formula usage

Video:

[Using Formulas](#)

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4.3: Video Links

1. [Selecting The Formula](#)
 2. [Triangle Method Lecture](#)
 3. [Using Formulas](#)
 4. [Using a triangle to rearrange simple formulas](#)
-

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4.4: Evaluation

If you are interested in the evaluation files, they can be found at the following links

▮ [Applying Common Formulas](#)

▮ [Formula Usage](#)

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5.1: Video Links

[Word Problem Lecture Part 1](#)

[Word Problem Lecture Part 2](#)

[5 steps to solve word problems](#)

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5.2: Useful Tool Links

[Useful link for hydraulic formulas](#)

[On line cylinder calculator tool](#)

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6.1: Video Links

[Tables Lecture Part 1](#)

[Tables Lecture Part 2](#)

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6.2: Evaluation

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